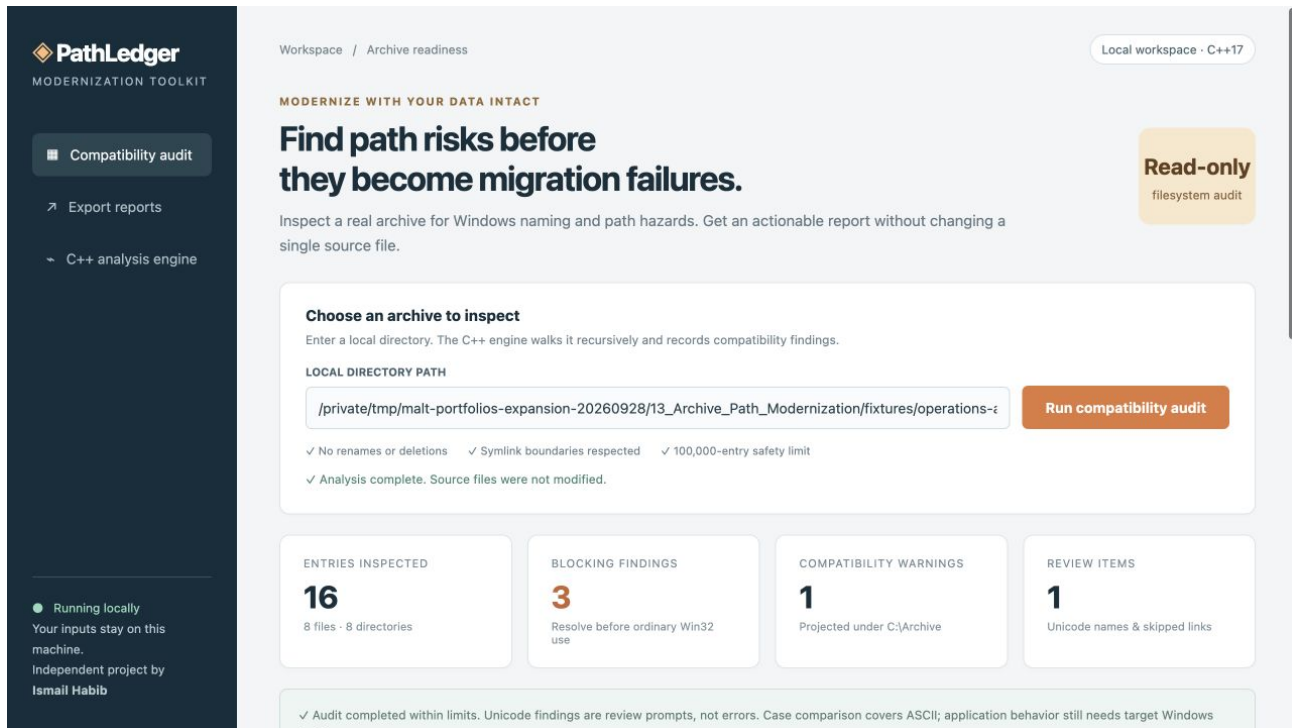


PathLedger

Windows Archive Compatibility Audit

Independent software project | Owner: Ismail Habib | Created September 2026



Actual application capture. Any example records are labeled verification inputs.

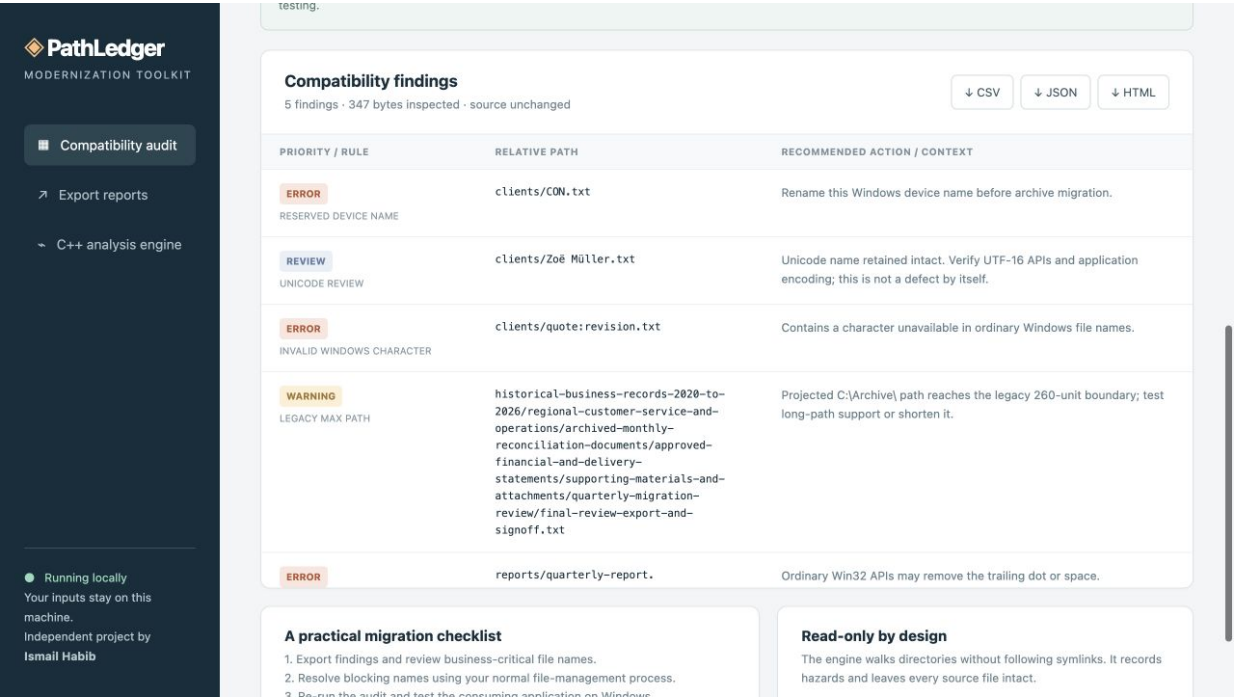
The problem

Archive migrations can fail when file names, path lengths and application encoding assumptions differ between older software and a newer Windows environment.

What the project delivers

- Recursive filesystem audit with symlink boundaries and scan limits
- Windows reserved names, invalid characters, trailing dots/spaces and ASCII case/trim conflicts
- Projected legacy path-length checks and explicit Unicode review findings
- Local dashboard with downloadable JSON, CSV and standalone HTML reports

Built to be inspected.



Actual compiled-engine findings from the labeled synthetic archive, with names, rules and next-step guidance.

What was verified

Compiled with Apple clang++ in C++17 mode with no warnings. Nine engine tests and six live HTTP checks passed. The bundled archive produced 3 blocking findings, 1 path warning and 1 Unicode review item.

How it works

Real local directory to C++17 read-only enumeration to compatibility rules to review dashboard to JSON / CSV / HTML report

Current scope

Independent new modernization-support tool; no historical client migration is claimed. Windows execution and MSVC compilation are not verified. ASCII case comparison is not full Windows Unicode case folding. Long-path checking assumes C:\Archive\; application-specific Windows validation remains required.

Included: source code, setup guide, tests, screenshots, editable case study and Malt copy.
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